Analysis of Failure and repair data

There are two general approaches to fitting reliability distributions to failure data. The first and usually preferred, method is to fit a theoretical distribution, such as the exponential, Weibull, normal or lognormal distributions. The second is to derive, directly from the data, an empirical reliability function or hazard rate function.

Sources of data

1. Operational or field data reflecting normal use of the component.
2. Data generated through reliability testing and includes screening or burn-in testing, life or accelerated life testing and reliability growth testing.

CENSORING

A common problem in generating reliability data is censoring. It occurs when the data are incomplete because units are removed from consideration prior to their failure or because the test is completed prior to all units failing.

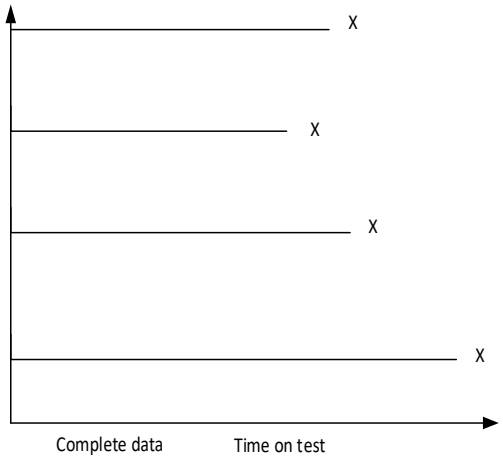
Singly censored data: All units have the same test time, and the test is concluded before all units have failed.

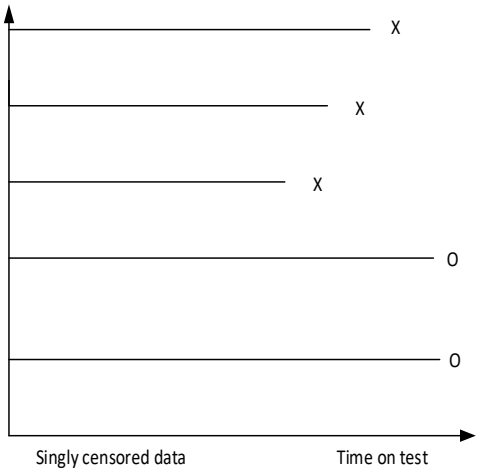
Type I censoring: Testing is terminated after a fixed length of the time, t^* , has elapsed.

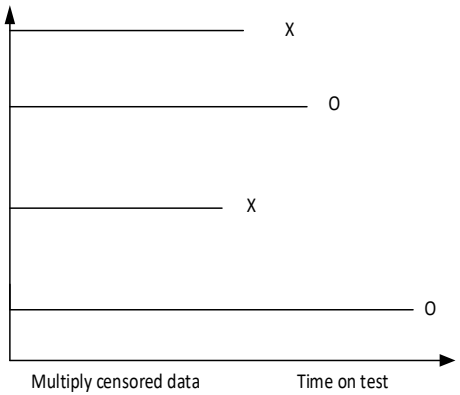
Type II censoring: Testing is terminated after a fixed number of failures, r, has occurred. The test time is then given by t_r, the failure time of the rth failure.

Multiply censored data: Test times or operating times differ among the censored units. Censored units are removed at various times from the sample.

Data not having any censored units are referred to as **complete data.**







Effect of censored data in the analysis:

Censoring introduces additional difficulties in the statistical analysis of failure times. To ignore censored units in the analysis would eliminate valuable information and would bias the results. For example, if the remaining operating units from Type-I testing were ignored, only the weakest units having the earliest failure times would be treated in the analysis and reliability of the component would be seriously under-estimated.

Empirical Methods:

Empirical methods of analysis are also referred to as **nonparametric** methods or **distribution free** methods. The objective is to derive, directly from the failure times, the failure distribution, reliability function and hazards rate function.

Ungrouped complete data

Given that $t_1, t_2, \ldots\ldots t_n$ where $t_i \leq t_{i+1}$ are n **ordered failure times** comprised in a random sample, the number of unit surviving at time t_i is $n - i$. Therefore, a possible estimate for reliability function,

$$\widehat{R}\,(\mathrm{t}_i) = \frac{n-i}{n} = 1 - \frac{i}{n}$$

$$\widehat{F}(\mathrm{t}_i) \;=\; 1 - \widehat{R}\,(\mathrm{t}_i) \;=\; \frac{i}{n}$$

An improved estimate of the cumulative failure distribution is $\widehat{F}(t_i) = \frac{i}{n+1}$ This implies that an equal number of failures will occur in the intervals $(0, t_1), (t_1, t_2), \ldots\ldots\ldots\ldots, (t_{n-1}, t_n), (t_n, t_\propto)$.

$$\widehat{R}\,(\mathrm{t}_i) \;=\; 1 - \frac{i}{n+1} = \frac{n+1-i}{n+1}$$

$\hat{F}(t)$ estimates are sometimes referred to as **plotting positions** since they provide the ordinate values in plotting the cumulative distribution function. That is, the points $\left(t_i, \hat{F}(t_i)\right)$ provide a graph of the estimate of $F(t)$.

Equation $\hat{F}(t_i) = \frac{i}{n+1}$ provides the mean plotting position for i^{th} ordered failure. An alternative plotting position is based on the median. The formula $\hat{F}(t_i) = \frac{i-0.3}{n+0.4}$ (Benard's approximation) is often used as an approximation of median position. For relatively large sample sizes, the differences among plotting positions $\frac{i}{n+1}$ and $\frac{i-0.3}{n+0.4}$ are insignificant.

Probability density function and hazards rate function

$$\hat{f}(t) = -\frac{\hat{R}(t_{i+1}) - \hat{R}(t_i)}{t_{i+1} - t_i} \qquad for\ t_i < t < t_{i+1}$$

$$= -\frac{\frac{n+1-i-1}{n+1} - \frac{n+1-i}{n+1}}{t_{i+1} - t_i} \qquad for\ t_i < t < t_{i+1}$$

$$= \frac{1}{(t_{i+1} - t_i)(n+1)} \qquad for\ t_i < t < t_{i+1}$$

Therefore, $\hat{\lambda}(t) = \frac{\hat{f}(t_i)}{\hat{R}(t_i)} = \frac{1}{(t_{i+1} - t_i)(n+1-i)} \qquad for\ t_i < t < t_{i+1}$

An estimate of the mean time to failure is obtained directly from the sample mean

$$\widehat{MTTF} = \sum_{i=1}^{n} \frac{t_i}{n}$$

An estimate of the variance of the failure distribution may be obtained from the sample variance:

$$s^2 = \sum_{i=1}^{n} \frac{\left(t_i - \widehat{MTTF}\right)^2}{n-1}$$

$$s^2 = \frac{\sum_{i=1}^{n} {t_i}^2 - n\widehat{MTTF}^2}{n-1}$$

If the sample of n failure times is large, an approximate $100(1-\alpha)$ percent confidence interval for the underlying MTTF may be obtained using

$$\widehat{MTTF} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}$$

Where $t_{\alpha/2,n-1}$ is found from a table of values the student's t distribution based on $n-1$ degree of freedom and the desired confidence level $(1-\alpha)$.

Example: Given the following 10 failures times in hours; 24.5, 18.9, 54.7, 48.2, 20.1, 29.3, 15.4, 33.9, 72.0, 86.1. Estimate $R(t), F(t), f(t)$ and $\lambda(t)$ and compute a 90% confidence interval for the MTTF of these ungrouped data.

Grouped complete Data

Failure times that have been placed into intervals, their originals values no longer being retained, are referred to as grouped data. Since the individual observation are no longer available; let $n_1, n_2, n_3, \ldots . n_k$ be the number of units having survived at ordered times $t_1, t_2, t_3, \ldots . t_k$ respectively. The logical estimate for $R(t)$ is

$$\widehat{R}\,(t_i) = \frac{n_i}{n} \quad i = 1,2, \ldots . . k$$

Where n is the number of units at risk at the start of the test. Because of the large sample size of the group data, it is generally unnecessary to obtain more precise estimates by considering plotting positions.

$$\hat{f}(t) = -\frac{\hat{R}(t_{i+1}) - \hat{R}(t_i)}{t_{i+1} - t_i}$$

$$= \frac{n_i - n_{i+1}}{(t_{i+1} - t_i)n} \quad for\ t_i < t < t_{i+1}$$

and $\hat{\lambda}(t) = \frac{\hat{f}(t_i)}{\hat{R}(t_i)} = \frac{n_i - n_{i+1}}{(t_{i+1} - t_i)n_i} \quad for\ t_i < t < t_{i+1}$

The MTTF is estimated on the basis of the midpoint of each interval. That is,

$\widehat{MTTF} = \sum_{i=0}^{k-1} \overline{t_i} \times \frac{(n_i - n_{i+1})}{n}$ where $\overline{t_i} = \frac{t_i + t_{i+1}}{2}$ and $t_0 = 0, n_0 = 0$

and $\frac{(n_i - n_{i+1})}{n}$ is the fraction that failed in interval $i - i + 1$. The sample variance is

$$s^2 = \sum_{i=0}^{k-1} \overline{t_i}^2 \frac{(n_i - n_{i+1})}{n} - \widehat{MTTF}^2$$

Example: Seventy compressors are observed at 5- month intervals with the following number of failures: 3, 7, 8, 9, 13, 18 and 12. Estimate $R(t), f(t)\ and\ \lambda(t)$ and determine the sample mean time to failure and sample standard deviation.

A **censored observation** is defined as an observation with incomplete information about the 'time-to event'. Type of censoring may be **right censoring**, **left censoring** and **interval censoring**.

Right censoring: The information about time to event is incomplete because the subject did not have an event during the time when the subject was studied. Failure times for some units are known only to be after some specified time, i.e., actual failure time exceeds observation.

Left censoring: Failure times for some units are known to occur only before some specified time, i.e. actual life time less than observation. It may occur when the item is inspected at a fixed time point.

Interval censoring: Failure times are known only to lie within an interval instead of being observed exactly. Observation give upper and lower bound on failure time. It occurs often with scheduled inspections.

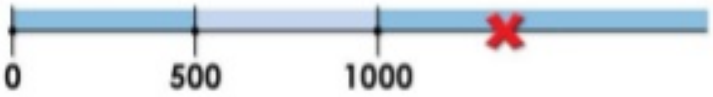
Left Censoring



Miles

TTF: $T < 8000$

Right Censoring



Hours

TTF: $T > 1000$

Interval Censoring



Miles

$9000 < TTF < 12{,}000$

Ungrouped Censored data:-

n units are placed in test and r failures occurring when, r < n. For data singly censored on the right the estimates of $R(t), f(t)$ and $\lambda(t)$ may be computed from expression developed for complete data.

$$\hat{R}(t_i) = \frac{n+1-i}{n+1},\ \hat{f}(t_i) = \frac{1}{(t_{i+1}-t_i)(n+1)}$$

$$\hat{\lambda}(t_i) = \frac{1}{(t_{i+1}-t_i)(n+1-i)}$$

The estimated reliability curve is truncated on the right at the time the test is terminated. In this case fitting a theoretical distribution may provide a more complete picture of the failure process in the right-hand tail of the distribution and allow MTTF to be computed.

For multiply censored data, t_i will represents a failure time and $\boldsymbol{t_i}^+$ **will represent a censored (removal) time**. The lifetime distribution of the censored units is assumed to be the same as that of uncensored unit. The sample consists of a set of ordered failure times plus censored times: $t_1, t_2, {t_3}^+, \ldots \ldots t_i,\ t_{i+1}^+, \ldots \ldots, t_n$. Three methods are used for estimating the reliability function:

1. **Product limit estimator**
2. **Kaplan- Meier form of product limit estimator**
3. **Rank adjustment method**

Product limit estimator:

It uses the $\hat{R}(t_i)$ expression developed for complete data.

$$\hat{R}(t_i) = \frac{n+1-i}{n+1}$$

Hence, $\hat{R}(t_{i-1}) = \frac{n+2-i}{n+1}$

$$\Rightarrow \frac{\hat{R}(t_i)}{\hat{R}(t_{i-1})} = \frac{n+1-i}{n+2-i}$$

$$\Rightarrow \hat{R}(t_i) = \frac{n+1-i}{n+2-i}\hat{R}(t_{i-1})$$

If the censoring rather than a failure takes place at time t_i, the reliability should not change and $\hat{R}(t_i^+) = \hat{R}(t_{i-1})$. We can generalize it

$$\delta_i = 1 \; if \; failure \; occurs \; at \; time \; t_i$$

$$0 \; if \; censoring \; occurs \; at \; time \; t_i$$

Then, $\hat{R}(t_i) = \left(\frac{n+1-i}{n+2-i}\right)^{\delta_i} \hat{R}(t_{i-1})$

When, $\hat{R}(t_0 = 0) = 1$

Therefore, $\hat{R}(t_i) = \begin{cases} \left(\frac{n+1-i}{n+2-i}\right)\hat{R}(t_{i-1}) & if \; observation \; i \; is \; a \; recorded \; failure \\ \hat{R}(t_{i-1}) & if \; observation \; i \; is \; a \; suspended \; item \end{cases}$

Examples:

The following failure and censor times (in operating hours) were recorded on 10 turbine vanes.

$150, 340^{+}, 560, 800, 1130^{+}, 1720, 2470^{+}, 4210^{+}, 5230, 6890.$

Censoring was a result of failure modes other than fatigue or wear out. Derive an empirical reliability curve. Calculate reliability of the turbine vanes by product limit estimator method.

Kaplan-Meier form of product limit estimator:-

$t_j = ordered\ failure\ times$

$n_j = number\ remaining\ at\ risk\ just\ prior\ to\ the\ j^{th}\ failure$

Assuming that there are no ties in failure times and the censoring times do not coincide with failure times, the Kaplan-Meier product limit estimator is given by:

$$\hat{R}(t) = \prod_{\{j:t_j \leq t\}} \left(1 - \frac{1}{n_j}\right) for\ 0 \leq t\ < t_1, R(t_0 = 0) = 1$$

Each term in this equation represent an estimate of the conditional probability of surviving past time t_j given survival just prior to time t_j.

J. F. Lawless discusses a number of the properties of the Kaplan-Meier product limit estimator and provide the following estimate of its variance.

$$\widehat{var}\left[\hat{R}(t)\right] = \hat{R}(t)^2 \sum_{\{j:t_j<t\}} \frac{1}{n_j\left(n_j-1\right)}$$

Example: The following failure and censor times (in operating hours) were recorded on 10 turbine vanes.

$150, 340^+, 560, 800, 1130^+, 1720, 2470^+, 4210^+, 5230, 6890.$

Using the above multiply censored data with $R(t_i + 0)$ representing the reliability immediately following the ith failure, compute an empirical reliability function by means of the Kaplan-Meier product limit estimator.

Rank Adjustment Method:

An alternative approach due to L G Johnson for estimating $\hat{F}(t_i)$ and $\hat{R}(t_i)$ with multiply censored data. This approach make use of $\hat{F}(t_i) = \frac{i-0.3}{n+0.4}$ for estimating the $\hat{F}(t_i)$ and $\hat{R}(t_i)$ with multiply censored data while adjusting the rank order, if necessary, of the ith failure to account for censored times occurring prior to the ith failure. Since a censored unit has some probability of failing before or after the next failure (or failures), it will influence the rank of subsequent failures. For example, suppose the following data are obtained (1) failure at 50hr (2) Censor at 80hr (3) failure at 160hr. Then the first failure will have rank 1; however, the second failure could have rank 2 if the censored unit fails after 160hr. or it could have rank 3 if the censored unit failed before 160hr.

Therefore, the second failed unit will be assigned a rank order between 2 and 3 on the basis of the following formula derived from considering all possible rank position of the censored unit.

$$Rank\ increment = \frac{(n+1) - i_{t_{i-1}}}{1 + number\ of\ units\ beyond\ present\ censored\ unit}$$

Where n is the total no. of units at risk and $i_{t_{i-1}}$ is the rank order of **failure time** $i-1$. The rank increment is recomputed for the next failure following a censored unit. Its adjusted rank then becomes

$$i_{t_i} = i_{t_{i-1}} + rank\ increment\ \ and$$

$$\hat{R}(t_i) = 1 - \frac{i_{t_i} - 0.3}{n + 0.4}$$

The rank increment then remains the same until the next censor takes place.

Example: The following failure and censor times (in operating hours) were recorded on 10 turbine vanes.

$$150, 340^{+}, 560, 800, 1130^{+}, 1720, 2470^{+}, 4210^{+}, 5230, 6890.$$

Using the above multiply censored data compute an empirical reliability function using the rank adjustment method.